

Vision Guard: AI-Powered Dry Eye Prediction and Prevention

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Abstract — Dry Eye Disease (DED) is a prevalent chronic condition that affects a significant portion of the population, particularly in Saudi Arabia, where up to 75% of individuals may be affected. DED arises from inadequate tear production, causing discomfort and visual disturbances. Contributing factors include hormonal changes, environmental conditions, and modern lifestyle habits such as excessive screen time, poor sleep, and stress. This project aims to develop an intelligent system leveraging machine learning techniques to assess DED risk based on lifestyle patterns. By using both supervised and unsupervised learning, the system identifies hidden risk factors and provides personalized recommendations for symptom management. The system also facilitates early detection, helping users take proactive measures to protect their eye health and improve overall quality of life.

1 INTRODUCTION

Dry Eye Disease (DED) is a growing concern worldwide, particularly in regions like Saudi Arabia, where it is estimated that up to 75% of the population is affected by this condition. DED occurs when there is insufficient tear production or poor tear quality, leading to discomfort and visual disturbances. The disease is exacerbated by several factors, including hormonal changes, environmental conditions, and modern lifestyle habits such as excessive screen time, poor sleep, and high stress levels. Left untreated, DED can result in severe complications such as corneal abrasions and potential vision loss, significantly affecting an individual's quality of life. As DED becomes more prevalent, it has become increasingly important to explore new methods of identifying and mitigating the risks associated with the condition.

The objective of this paper is to develop a machine learning-based system that assesses the risk of DED based on individual lifestyle factors. The system will employ both supervised and unsupervised learning techniques to identify hidden risk factors related to DED, including habits such as screen time, sleep quality, and environmental influences. Additionally, the paper aims to provide personalized recommendations to help mitigate symptoms of DED, enabling users to make better lifestyle choices. Early detection of DED through this system will raise awareness and allow users to take proactive measures to protect their eye health.

The paper is structured as follows: Section 2 discusses the background and related work in the field of DED and machine learning applications. Section 3 presents the State of the Art, highlighting recent machine learning studies related to Dry Eye Disease and outlining the baseline models we replicated for comparison. Section 4

explains the methodology employed in this study, including the data collection, preprocessing steps, and the machine learning models used. Section 5 presents the evaluation of the model. Section 6 provides a discussion of the findings and compares them to existing literature. Section 7 concludes the paper and suggests directions for future research. Finally, Section 8 provides a comprehensive list of all references cited in this study.

2 LITERATURE REVIEW

Machine learning has demonstrated encouraging outcomes in forecasting various chronic illnesses by utilizing only behavioral and lifestyle information, thereby reducing the necessity for clinical data in certain instances. Nevertheless, this methodology has not been extensively examined in relation to Dry Eye Disease (DED), where the majority of current research continues to depend significantly on clinical metrics such as tear composition and imaging of the meibomian glands.

Our research aims to assess whether behavioral trends can yield sufficient information to predict the risk of DED without the inclusion of clinical data. To facilitate this exploration, we will review previous studies across three categories: those employing lifestyle data for other health conditions, those utilizing clinical data specifically for DED, and studies that integrate both types of data.

Several studies, summarized in **Table 2.1** and detailed in Appendix A, have explored the potential of behavioral and lifestyle data to predict chronic diseases without relying on clinical metrics.

- Study 1 (2022): Applied ML techniques to predict chronic kidney disease using only lifestyle and behavioral data. It used LightGBM and achieved an AUC of 0.71.
- Study 2 (2022): Focused on Type II Diabetes Mellitus prediction using lifestyle indicators and reported an accuracy of 99.41% with F1-score values of 99.15%.

These studies, although targeting different conditions, highlight the untapped potential of lifestyle data in healthcare prediction raising the question of whether a similar approach could be successfully applied to DED.

Table 2.2 summarizes studies that rely exclusively on clinical data for predicting Dry Eye Disease (DED); full details for each study are available in Appendix B.

Clinical data such as tear composition, gland imaging, and ocular surface factors are the main focus of the majority of current research on DED prediction. Although behavioral and lifestyle factors are frequently left out of these research, which may restrict their generalizability and early intervention potential, they do show the diagnostic value of clinical inputs.

- Study 3 (2024): Utilized infrared meibography images to assess Meibomian gland morphology and its correlation with DED symptoms. The accuracy for morphological characteristics ranged from 65% to 99%.
- Study 5 (2024): Focused on classifying cataract patients based on their DED diagnosis. The study utilized tear samples and reported an accuracy of 97.54% with ROC curve values of 0.99.

Despite their great accuracy, these techniques rely on clinical tests, which are frequently expensive, intrusive, or prohibitive for routine monitoring. More significantly, they ignore lifestyle risk factors that could be used as a preventative measure that may contribute to the early development of DED. Our work aims to close this gap by concentrating on behavior-based, non-invasive prediction.

Table 2.3 provides a summary of studies that combine both lifestyle and clinical data in the context of DED prediction. Full study details are presented in Appendix C.

The integration of lifestyle and clinical data for DED prediction has been investigated in a smaller number of studies. Although these hybrid techniques provide a more comprehensive perspective, they frequently place more emphasis on clinical factors, and few of them offer insight into how lifestyle modifications could lessen the symptoms of DED.

- Study 9 (2024): Utilized lifestyle and clinical data to predict DED symptoms and diagnoses. It employed logistic regression and reported accuracy ranges from 60.7% to 98.7% for different DED indicators.
- Study 10 (2020): Integrated lifestyle factors from the Korea National Health and Nutrition Examination Survey (KNHANES) and used decision trees to identify significant DED risk factors like dietary habits and smoking. The AUROC score achieved was 0.70, with specificity and sensitivity values ranging from 66% to 68%.

Although the majority of these research still rely significantly on clinical inputs, they demonstrate the benefits of using lifestyle data. Furthermore, not many research assess if lifestyle data by itself can produce prediction performance that is equivalent. The goal of our work is to identify and evaluate the prediction power of lifestyle factors alone, which is still a relatively untapped area in DED research.

Gaps in Current Solutions

While existing research has provided valuable insights into DED prediction, there are notable gaps:

- **Overemphasis on Clinical Data:** Many studies rely heavily on clinical data, missing the potential influence of lifestyle factors.
- **Limited Use of Unsupervised Learning:** Most studies focus on supervised learning, missing the opportunity to uncover hidden patterns in lifestyle data.
- **Lack of Personalized Recommendations:** Existing models typically predict DED risk without providing personalized lifestyle recommendations for prevention or management.

Our project addresses these gaps by:

- **Focusing on Lifestyle Data:** We leverage a comprehensive dataset that includes variables like sleep quality, screen time, and eye strain, which are critical for understanding DED.
- **Integrating Unsupervised Learning:** We utilize unsupervised learning to discover hidden patterns in lifestyle data, offering deeper insights into DED risk factors.
- **Developing a Personalized Recommendation System:** Unlike existing research, our approach includes a system that predicts DED risk and provides real-time, tailored recommendations to help users improve their lifestyle and reduce DED symptoms.

In summary, the literature currently in publication shows that clinical data is heavily relied upon for DED prediction, but lifestyle-based approaches are still neglected and poorly researched. The function of behavioral data in DED is yet unknown, despite the fact that it has been shown to be useful in predicting other chronic illnesses. By assessing whether DED can be reliably predicted using simply lifestyle data, as well as if such models can facilitate early detection and provide individualized, non-invasive care solutions, this study seeks to close this gap.

3 STATE OF THE ART

To establish a solid foundation for our classification approach, we replicated three state-of-the-art machine learning studies related to Dry Eye Disease (DED) detection using both behavioral and biological data.

The first replication was based on a study that utilized lifestyle and behavioral factors such as screen time, stress level, and sleep quality to predict DED symptoms using Logistic Regression. We engineered relevant features, including Body Mass Index (BMI) and interaction terms (e.g., stress $\times$ sleep quality), and addressed class imbalance using SMOTE. Although

Logistic Regression assumes linearity, incorporating interaction features enhanced its expressiveness. Our implementation achieved a test AUC of 0.77 and an accuracy of 73%.

The second replication employed LightGBM, inspired by a study that predicted chronic kidney disease using similar lifestyle data. Following a structured pipeline of preprocessing, feature selection, SMOTE, and hyperparameter optimization, our model yielded

improved performance (AUC = 0.78, Accuracy = 76%) with high recall for DED-positive cases. The primary challenge was adapting LightGBM which was originally trained on a large and balanced dataset to our smaller, behavior-only dataset. This was addressed through hyperparameter tuning and stratified data handling.

The third replication followed a metabolomics-based study that compared eight machine learning algorithms for classifying DED patients. Logistic Regression with L2 regularization was the best-performing model. We replicated the study's methodology by applying $\log_2$ transformation, quantile normalization, and nested cross-validation using GridSearchCV. While our AUC and sensitivity scores were comparable to the original study, slight reductions in specificity and MCC were observed, likely due to differences in sample size and preprocessing. Challenges included accurately implementing the normalization techniques, managing the computational load of nested validation, and addressing class imbalance due to the limited number of control samples. These were mitigated by precisely applying the same transformations, using stratified cross-validation, and setting `class_weight='balanced'`.

Across all three replications, we encountered common challenges such as class imbalance, data limitations, and model assumptions. Through rigorous preprocessing, feature engineering, and robust validation, we achieved reliable and competitive performance aligned with the standards of existing research.

4 METHODOLOGY

4.1 Dataset, EDA and Pre-Processing

4.1.1 Dataset Description:

The dataset represents a comprehensive collection of information related to Dry Eye Disease (DED) with 20,000 records and 26 features. The features are a mix of numerical and categorical variables. The dataset includes the following key categories:

- Demographics: Gender, Age, Height, Weight
- Lifestyle Habits: Screen time, Sleep quality, Caffeine/alcohol consumption
- Health Indicators: Heart rate, Blood pressure, [medical issues](#)
- Environmental Factors: Smart device use before bed, Blue-light filter usage
- Eye Health Symptoms: Eye strain, Redness, Irritation

-Target Variable: Dry Eye Disease (Yes/No)

Source:

This dataset was sourced from Kaggle and consists of self-reported data collected through surveys or questionnaires about Dry Eye Disease (DED).

4.1.2 Exploratory Data Analysis (EDA)

4.1.2.1 Class Label:

The target variable, Dry Eye Disease (DED), exhibits class imbalance, with:

-13,037 samples labeled "Y" (Yes)

-6,963 samples labeled "N" (No)

This imbalance may affect the model's performance, as the model may be biased toward predicting the majority class. The class imbalance will be addressed during the pre-processing phase.

4.1.2.2 Handling Missing Data:

There were no missing or null values across any of the features, ensuring data completeness. Additionally, no duplicate records were found, contributing to the integrity and reliability of the data for modeling.

4.1.2.3 Outliers:

The dataset was analyzed for outliers using the Interquartile Range (IQR) method. This revealed that no features contained significant outliers, ensuring a clean and well-structured dataset.

4.1.2.4 Correlation Analysis:

A correlation heatmap was generated to assess the relationships between features. Categorical features were temporarily encoded (e.g., 'Yes'/'No', 'Male'/'Female') to numeric values. The analysis showed that there is minimal feature redundancy and weak linear correlations (ranging between -0.03 and +0.03) among most features, reducing the risk of multicollinearity.

4.1.3 Pre-Processing

4.1.3.1 Data Cleaning:

-Missing Values: No missing or null values were found.

-Duplicate Records: No duplicates were detected.

-Categorical Formatting: Categorical features were standardized for consistency (e.g., "y" → "Y").

4.1.3.2 Categorical Encoding:

Label Encoding was used for categorical variables with binary or gender-based categories to ensure efficient and compact modeling. Categorical features were mapped to integers to prepare them for machine learning models.

4.1.3.3 Feature Engineering:

The composite 'Blood pressure' column was split into two numerical features, Systolic_BP and Diastolic_BP, and the original column was removed for clarity.

4.1.3.4 Feature Scaling (Normalization):

Min-Max Scaling was applied to continuous features such as Sleep Duration and Screen Time to normalize

the data to a range of [0, 1], ensuring compatibility with distance-based models.

4.1.3.5 Handling Class Imbalance

SMOTE (Synthetic Minority Over-sampling Technique):

Given the significant class imbalance (more instances of "Yes" than "No"), SMOTE was applied to the training set to balance the class distribution by generating synthetic instances of the minority class. SMOTE helps prevent overfitting, improves model generalization, and ensures that both classes are represented in the model's training process.

4.2 Modeling

4.2.1 Unsupervised Learning: Clustering

In the unsupervised part of this project, we implemented and compared four clustering algorithms: K-Means, DBSCAN, Agglomerative Clustering, and Gaussian Mixture Models (GMM). These models were selected because they represent a diverse range of clustering approaches: partitioning (K-Means), density-based (DBSCAN), hierarchical (Agglomerative), and probabilistic (GMM), allowing us to comprehensively explore the dataset's underlying structure.

We tuned the parameters of each model to optimize performance. For K-Means, we used the elbow method and silhouette scores to choose the optimal number of clusters $k=5$. For DBSCAN, we tested multiple values of ϵ (0.5–2) and min_samples (3, 5, 10), identifying the combination that produced the most meaningful clusters. In Agglomerative Clustering, we applied Ward's linkage and determined the natural cut height by examining the largest gap in the dendrogram, resulting in four clusters. For GMM, we evaluated model fits using BIC and AIC scores to select the best number of components.

Feature selection was performed using a K-Nearest Neighbors (KNN) graph and Laplacian scoring to rank the importance of features. This process identified the six most relevant variables — sleep duration, sleep quality, stress level, average screen time, smart device use before bed, and blue-light filter usage — which were then used as input for clustering.

The primary goal of applying unsupervised learning was to uncover natural groupings in the data that reflect lifestyle behaviors and physiological characteristics linked to eye health, without relying on the known Dry Eye Disease labels. This exploratory approach aimed to

reveal hidden patterns and potentially identify subgroups at higher risk, providing insights that could inform personalized health interventions.

4.2.2 Supervised Learning: Clustering

To predict the presence of Dry Eye Disease (DED), we implemented and evaluated three supervised machine learning algorithms: Logistic Regression, XGBoost, and LightGBM. These models were selected based on a thorough review of recent literature, where they demonstrated strong predictive performance in health-related classification problems using behavioral and lifestyle features. Their proven effectiveness, flexibility, and scalability made them suitable for our project objectives.

Logistic Regression was chosen for its interpretability and widespread use in binary classification tasks.

XGBoost was selected due to its robustness in handling high-dimensional structured data and its gradient boosting framework.

LightGBM was implemented for its high computational efficiency and superior accuracy on large-scale datasets, especially when dealing with categorical features.

All models were trained on the preprocessed dataset following the application of SMOTE to address class imbalance. Model evaluation was conducted using standard metrics including accuracy, precision, recall, F1-score, and AUC-ROC. Additionally, hyperparameter tuning was performed using GridSearchCV to optimize each model's performance.

4.2.3 Unsupervised Learning: Autoencoder

As part of our exploration of unsupervised learning methods, we initially applied clustering algorithms (e.g., K-Means, DBSCAN, GMM) to discover hidden structures in the lifestyle-related Dry Eye Disease (DED) dataset. However, these models demonstrated weak performance, with a maximum Silhouette Score of only 0.30. This result indicated poor separation and low-quality clustering, suggesting that the behavioral features lacked strong natural groupings.

To address this, we explored the use of Autoencoders—neural networks trained to learn efficient, compressed representations of high-dimensional data without relying on labeled outputs. Autoencoders are particularly suitable for extracting meaningful low-dimensional embeddings that can reveal latent structure.

Architecture and Training:

- **Encoder:** Two fully connected layers with 16 and 12 units, using ReLU activation and dropout to prevent overfitting.
- **Latent Bottleneck:** A compact 5-dimensional layer capturing the encoded representation of the data.

- **Decoder:** A mirrored structure reconstructing the original input using a sigmoid activation at the output layer to ensure values remain in the $[0, 1]$ range—appropriate for the normalized feature space which included some negative values.

4.2.4 Supervised Learning: Autoencoder

After obtaining the compressed features from the bottleneck layer of the Autoencoder, we evaluated their utility in supervised classification. Specifically, we assessed whether these unsupervised representations retained sufficient discriminatory power to predict Dry Eye Disease.

The compressed features were fed into three classifiers: **Logistic Regression, XGBoost, LightGBM.**

The classification pipeline followed standard practices:

- The dataset was split into training and test sets
- SMOTE was applied to balance class distribution
- Hyperparameter tuning was performed using GridSearchCV for each model

This supervised learning phase enabled us to directly compare performance using Autoencoder features versus original features and those derived from clustering.

5 EVALUATION

5.1 Experimental Design

5.1.1 Unsupervised Learning: Clustering

To evaluate the performance of the unsupervised clustering models, we designed a structured experimental framework.

First, we drew a stratified random sample of 10,000 records from the complete dataset, ensuring that the distribution of the target label (Dry Eye Disease) was preserved. This sampling approach allowed us to work with a manageable yet representative subset while maintaining diversity across relevant subgroups.

For feature selection, we initially considered a broad set of physiological and lifestyle variables, including sleep duration, sleep quality, stress level, average screen time, smart device usage before bed, blue-light filter use, heart rate, physical activity, and blood pressure measures. To systematically identify the most informative features, we constructed a K-Nearest Neighbors (KNN) graph (with $k=5$), computed the associated graph Laplacian, and ranked the features using Laplacian scores, which prioritize variables that best preserve local neighborhood structures. Based on this process, we selected the top six features:

sleep duration, sleep quality, stress level, average screen time, smart device before bed, and blue-light filter use.

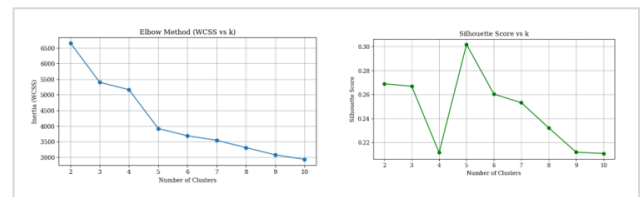
We then evaluated and compared multiple clustering algorithms, including **K-Means, DBSCAN, Agglomerative Clustering, and Gaussian Mixture**

Models (GMM). Each model's performance was assessed using established internal validation metrics:

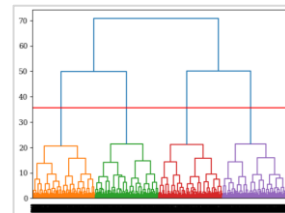
- **Silhouette Score**, reflecting the cohesion and separation of clusters.
- **Calinski-Harabasz Index**, measuring the ratio of between-cluster to within-cluster dispersion.
- **Davies-Bouldin Index**, quantifying intra-cluster similarity and inter-cluster separation (lower values indicate better performance).

For **parameter tuning**, we systematically explored different configurations:

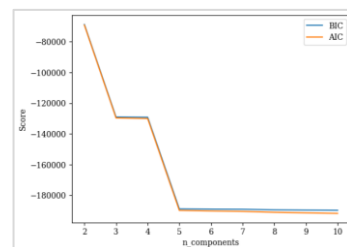
- For K-Means, we varied the number of clusters (k) from 2 to 10, using the elbow method and silhouette analysis to identify the optimal cluster count.



- For DBSCAN, we tuned the ϵ parameter (0.5, 1, 1.5, 2) and minimum samples (`min_samples`: 3, 5, 10) to optimize cluster density and stability.
- For Agglomerative Clustering, we analyzed the dendrogram's largest vertical gap to determine the most natural cutting height.

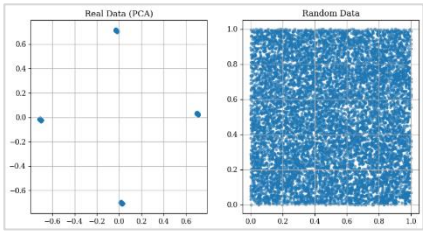


- For GMM, we evaluated models with 2 to 10 components, selecting the most parsimonious solution using the **Bayesian Information Criterion (BIC)** and **Akaike Information Criterion (AIC)**.



To support visual inspection, we applied Principal Component Analysis (PCA) to project the high-dimensional feature space onto two dimensions and generated cluster visualizations. These plots provided qualitative insights into the distinctness and compactness of the clusters formed by each model.

Finally, to assess the overall clusterability of the data, we computed the Kullback-Leibler (KL) divergence between the distribution of real data projections and synthetic random data. The observed KL divergence value (~ 3.24) indicated a moderate clustering tendency, supporting the feasibility of applying unsupervised methods to discover meaningful subgroup structures.



5.1.2 Supervised Learning: Classification

The experimental setup followed a consistent pipeline to ensure fair model comparison. The dataset was split into 80% training, 10% validation, and 10% testing using stratified sampling to maintain class balance.

To handle class imbalance, SMOTE was applied to the training data only. Each model was trained first with default settings, followed by hyperparameter tuning using GridSearchCV with 5-fold cross-validation based on model-specific parameters.

All models were evaluated using standard metrics including accuracy, precision, recall, F1 score, and AUC ROC on both validation and test sets. Confusion matrices and ROC curves were also plotted to assess class separation. This unified design ensured reliable and unbiased comparison across all models.

5.1.3 Unsupervised Learning: Autoencoder

To explore the potential of Autoencoders in extracting informative representations from lifestyle data, we trained an unsupervised **Autoencoder** using **Mean Squared Error (MSE)** loss over **300 epochs** with **early stopping** based on validation performance. A total of **8,000 stratified samples** (excluding the DED label) were used for training. The goal was to reduce dimensionality while preserving the informative aspects of the lifestyle features.

The model exhibited smooth convergence behavior, as evidenced by steadily declining training and validation loss curves.

The final **reconstruction error** averaged approximately **0.0981**, and loss trends indicated consistent learning with **no signs of overfitting**. These results confirm that the Autoencoder effectively compressed and reconstructed the input data, preserving the core statistical properties across most samples.

5.1.4 Supervised Learning: Autoencoder

Following the unsupervised training of the Autoencoder, we evaluated whether the compressed latent features retained sufficient class-discriminative information for

Dry Eye Disease (DED) prediction. The encoded representations were used as input to train three supervised classifiers.

The dataset was **split** into training and testing sets, and class imbalance was addressed using **SMOTE**. Each model underwent hyperparameter tuning via **GridSearchCV** to optimize performance.

However, the classification results using the compressed features were consistently lower compared to models trained on the original dataset and applying clustering. Specifically:

- **F1 Score** and **AUC-ROC** declined across all models.
- **AUC-ROC** values ranged between **0.56** and **0.60**, indicating weak separation between DED-positive and negative classes.

5.2 Results

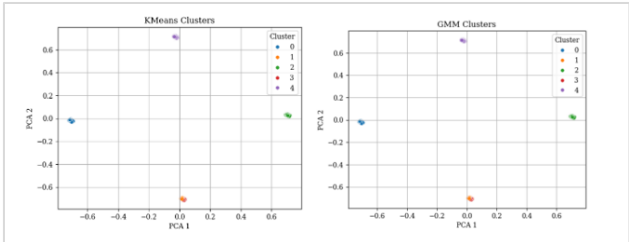
5.2.1 Unsupervised Learning: Clustering

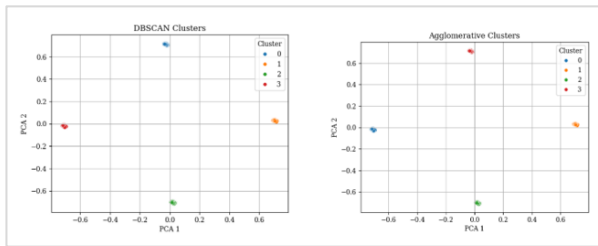
Model	Number of Clusters	Silhouette Score	Calinski-Harabasz Index	Davies-Bouldin Index
K-Means	5	0.3018	3339.56	1.466
DBSCAN	4	0.3545	4009.59	1.254
Agglomerative	4	0.3545	4009.59	1.254
GMM	5	0.3017	3337.90	1.468

The **DBSCAN** and **Agglomerative Clustering** models outperformed K-Means and GMM in terms of both the **silhouette score** (0.354 vs. ~ 0.302) and the **Davies-Bouldin index** (1.254 vs. ~ 1.466), indicating superior cluster compactness and separation.

Notably, DBSCAN and Agglomerative both identified **four distinct clusters**, compared to five clusters detected by K-Means and GMM.

To complement the quantitative evaluation metrics, we generated PCA visualizations for each clustering model, providing a two-dimensional projection of the clustered data. Although the silhouette scores for DBSCAN and Agglomerative clustering were moderate (around 0.35), the PCA plots visually revealed relatively clearer and more distinct groupings compared to K-Means and GMM, which showed more overlap between clusters. This visual inspection suggests that despite only moderate quantitative scores, DBSCAN and Agglomerative models may better capture meaningful lifestyle-related subgroups within the data, offering potential insights into behavioral patterns linked to eye health.



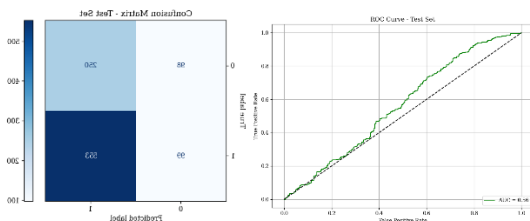


To enhance the supervised learning stage, we incorporated the cluster labels generated by the unsupervised DBSCAN model as an additional feature in the dataset. By appending the DBSCAN_Cluster column alongside the original features, we provided the supervised models with richer, pre-grouped structural information. This integration aimed to improve the model's ability to detect complex patterns and relationships, leveraging the unsupervised insights to boost classification performance on the target variable, *Dry Eye Disease*.

5.2.2 Supervised Learning: Clustering

Results – Logistic Regression The model was first trained with default hyperparameters on SMOTE-balanced data, achieving 60.40% accuracy, F1 score of 67.11%, and AUC-ROC of 63.36% on the validation set. After hyperparameter tuning using GridSearchCV, performance improved to 68.10% accuracy, F1 = 77.83%, and recall = 85.89%.

However, when retrained on combined training and validation data, performance dropped to 58.47% accuracy, suggesting possible overfitting. On the test set, the model reached 65.10% accuracy, F1 = 76.01%, but a lower AUC-ROC of 56.03%, indicating limited class separation.

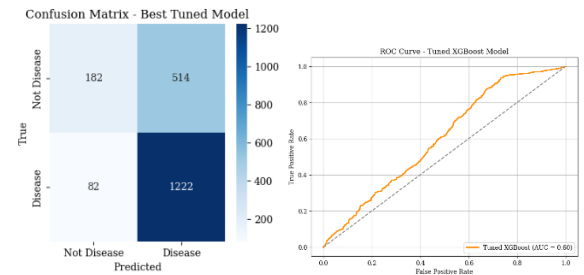


Results – XGBoost The initial XGBoost model, trained using default hyperparameters, achieved an accuracy of 64.20%, F1 score of 62.21%, precision of 61.83%, and recall of 64.20%. The confusion matrix revealed a considerable number of false positives and false negatives, limiting the model's generalization capability.

After applying GridSearchCV over 2,187 hyperparameter combinations, the best-tuned model significantly improved across all metrics: Accuracy: 70.20%, Precision: 69.89%, Recall: 70.20%, F1 Score: 65.61%, AUC-ROC: 0.60

The updated confusion matrix demonstrated a notable reduction in false negatives, and the model correctly identified more positive cases. However, the ROC curve indicates that the model's ability to separate the two

classes probabilistically remains modest, with an AUC of 0.60.

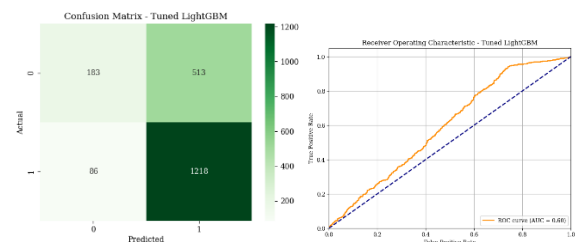


Results – LightGBM The LightGBM model, using default hyperparameters, achieved a solid test accuracy of 68.30%, with a strong recall of 89% and an F1 score of 79%. While the precision was moderate, the model demonstrated a strong ability to correctly identify positive cases of Dry Eye Disease, which is crucial in medical diagnostics.

After applying GridSearchCV for hyperparameter tuning, performance further improved to:

Accuracy: 70.05% | Precision: 70% | Recall: 93.4% | F1 Score: 80.26% | AUC-ROC: 0.60

The tuned model significantly enhanced recall while maintaining balanced accuracy and robustness. The updated confusion matrix confirms the model's improved sensitivity and reduced false negatives, aligning well with the project's goal of maximizing disease detection.



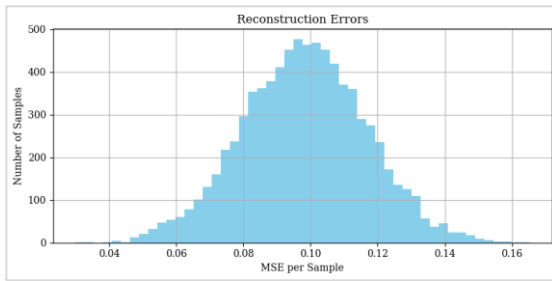
5.2.3 Unsupervised Learning: Autoencoder

The Autoencoder showed stable training behavior and achieved a low mean reconstruction error:

- **low Reconstruction Error (Mean):** 0.0981

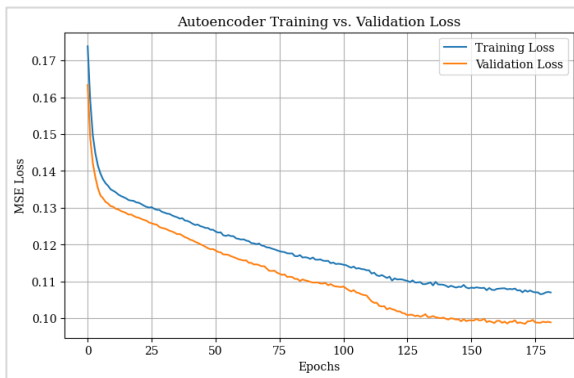
- **Training Dynamics:** Gradual decrease in both training and validation loss .

As shown in **Figure 1 (Reconstruction Errors)**, the majority of samples had an MSE close to the mean, with no major outliers, indicating consistent reconstruction performance across the dataset.



Loss curves **Figure 2 (Training vs. Validation Loss)**, showed smooth convergence and no signs of overfitting, confirming that the Autoencoder successfully learned a generalized, efficient compressed representation of the input data.

However, while the Autoencoder performed well as a compression model, its representations lacked the class-separability needed for downstream tasks.



5.2.4 Supervised Learning: Autoencoder

Model Performance on Reconstructed Sample Dataset:

Despite this, the compressed features performed poorly in classification tasks, where The same three classification models—**Logistic Regression**, **XGBoost**, and **LightGBM**—were trained and evaluated on a reconstructed sample dataset that was previously passed through an autoencoder. After splitting the data into training and testing sets and applying **SMOTE** to address class imbalance, each model underwent hyperparameter tuning via **GridSearchCV**.

The following table summarizes the optimal hyperparameters for each model, as selected by GridSearchCV:

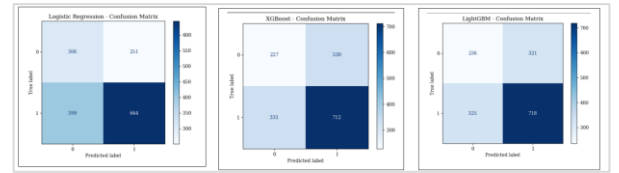
MODEL	BEST HYPERPARAMETERS
LOGISTIC REGRESSION	C=0.01, penalty="l2", solver="liblinear", fit_intercept=False, class_weight="balanced"
XGBOOST	learning_rate=0.1, max_depth=9, n_estimators=200, subsample=0.7
LIGHTGBM	learning_rate=0.1, max_depth=7, n_estimators=200, num_leaves=70

performance Summary on Sample Dataset (Test Set):

MODEL	ACCURACY	PRECISION	RECALL	F1-SCORE
LOGISTIC REGRESSION	0.59	0.72	0.62	0.66
XGBOOST	0.59	0.68	0.68	0.68
LIGHTGBM	0.60	0.69	0.69	0.69

Among the three, **LightGBM** emerged as the best-performing model, achieving the highest accuracy (60%)

and balanced classification metrics, particularly for the majority class (DED-positive).

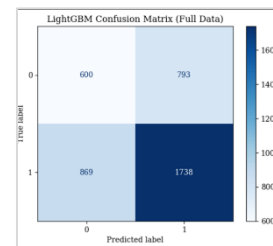


Evaluation of Full Reconstructed Dataset:

To validate model performance beyond the sampled dataset, we applied the trained autoencoder to the **entire DED dataset** and retrained the best model (LightGBM) using the same SMOTE technique for balance.

Final Test Results (Full Dataset):

(Accuracy: 58.45%, Precision: 68.67%, Recall: 66.67%, F1 Score: 67.65%)



These results indicate that, even after applying an alternative representation learning approach(e.g., **Autoencoder**) the model's performance remained suboptimal. Despite LightGBM showing relatively

better results, the drop in accuracy and recall—particularly on the full dataset—highlights a deeper issue.

This suggests that the core limitation lies in the dataset itself, where weak correlations and lack of distinctive patterns make it challenging for both unsupervised and supervised models to perform reliably.

6 DISCUSSIONS

Personalized Recommendations -- To enhance the practical value of the identified lifestyle clusters, we further analyzed the average behavioral profiles for each cluster derived from the unsupervised DBSCAN model. By grouping the dataset based on cluster assignments and calculating the mean values of key features (e.g., sleep duration, stress level, screen time), we created distinct lifestyle profiles that summarize the unique behavioral patterns within each subgroup.

To translate these profiles into actionable insights, we integrated an external language model (GPT-4) through the OpenAI API. For each cluster, we dynamically generated a prompt that included its summarized behavioral characteristics and requested personalized health and wellness recommendations. The language model produced clear, supportive

lifestyle advice tailored to each group’s specific profile, covering aspects such as sleep hygiene, stress management, and device usage habits.

This approach demonstrates the potential of combining unsupervised learning with generative AI to move beyond static analysis, offering interpretable and user-friendly outputs that can inform personalized intervention strategies or public health recommendations.

Discussion – Logistic Regression While the model showed reasonable performance, its linear nature limited its ability to capture the non-linear patterns in the data. SMOTE improved recall but increased false positives, reducing specificity.

The performance drop after tuning suggests overfitting to the validation set, which is common with simpler models. Overall, Logistic Regression served as a solid baseline but was outperformed by more advanced models like XGBoost and LightGBM.

Discussion – LightGBM LightGBM demonstrated the highest recall and F1 score among all tested models, making it highly suitable for sensitive detection tasks such as medical classification. Its ability to capture non-linear interactions contributed to its superior performance, especially in detecting positive disease cases.

Although the AUC-ROC value remained moderate at **0.60**, the model’s classification results were consistently strong. The slight trade-off in precision is acceptable given the high recall — a favorable

balance in clinical contexts where missing true cases is more critical than over-identifying.

In summary, LightGBM proved to be the most effective supervised model in this study, combining high sensitivity with robust overall performance.

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Logistic Regression	65.10%	68.87%	84.82%	76.01%	56.03%
XGBoost	70.20%	69.89%	70.20%	65.61%	60.00%
LightGBM	70.05%	70.00%	93.40%	80.26%	59.71%

Table 6.1: Comparative performance of supervised models on the test set

Model	Accuracy	F1 Score	Recall	AUC	Precision
Logistic Regression (SOA v1)	73%	78%	93%	77%	67%
Logistic Regression (SOA v2)	75%	80%	91%	81%	76%
LightGBM (SOA)	76%	79%	93%	78%	69%
XGBoost (SOA)	75.4%	79%	91%	77%	69%

Logistic Regression (Post-Unsupervised)	65%	76%	84.8%	56%	68.8%
LightGBM (Post-Unsupervised)	70%	80%	93%	60%	70%
XGBoost (Post-Unsupervised)	70%	65.6%	70%	60%	69.9%

Table 6.2: Through comparative analysis, it was observed that the state-of-the-art (SOA) models outperformed the supervised models trained after the application of unsupervised learning. The SOA models, particularly LightGBM, demonstrated higher accuracy and AUC, indicating more robust performance and stronger class separability.

One possible explanation for this performance gap is that the unsupervised learning phase may have introduced noise or distorted the original feature space, especially in the presence of complex and high-dimensional behavioral data. This could have negatively impacted the effectiveness of subsequent supervised models, particularly those with linear assumptions such as Logistic Regression.

Although some post-unsupervised models—most notably LightGBM—achieved high recall and F1 scores, these improvements often came at the cost of reduced precision or lower AUC. This suggests that while clustering may enhance sensitivity to the positive class, it does not always contribute to balanced or generalizable performance.

In summary, while unsupervised learning can be a valuable preprocessing step, its integration should be carefully evaluated, as it may not always align well with the structure and nature of the dataset being modeled.

Discussion – XGBoost XGBoost outperformed Logistic Regression, particularly in terms of recall and balanced accuracy, due to its ability to model non-linear patterns within the data. The hyperparameter tuning contributed to more stable performance and improved sensitivity toward minority class detection.

Despite these gains, the model still exhibited a high number of false positives and a relatively low AUC. This suggests potential challenges in feature separability or remaining noise in the dataset. Nonetheless, XGBoost demonstrated a stronger capacity to learn complex patterns in the Dry Eye Disease dataset compared to linear models.

After a comprehensive evaluation of both supervised and unsupervised models, we determined that the most effective supervised model was **LightGBM**. This decision was based on its superior recall (93.40%) and F1 score (80.26%) across all tested models, as shown in Table 1. These metrics are particularly crucial in the context of medical applications like Dry Eye Disease detection, where the ability to correctly identify positive cases (sensitivity) is more important than minimizing false positives.

While Logistic Regression provided a solid baseline and XGBoost demonstrated robust performance with improved non-linear pattern capture, LightGBM's ensemble learning approach proved best at balancing sensitivity and overall classification quality. This makes it the preferred choice for applications where missing positive cases could lead to significant health impacts.

In summary, LightGBM, enriched with insights from the unsupervised clustering phase, was selected as the final model due to its strong recall, balanced performance, and its ability to leverage both labeled and unlabeled data insights effectively.

7 CONCLUSION

This study demonstrates the feasibility of predicting Dry Eye Disease (DED) using lifestyle-related data alone, without the need for clinical metrics. By applying both supervised and unsupervised machine learning techniques, we explored how behavioral patterns such as screen time, sleep quality, and stress levels relate to DED risk. The results showed that supervised models performed well in classifying DED, while unsupervised approaches like clustering and autoencoders struggled to identify meaningful groupings or predictive structures in our dataset. These findings suggest that although lifestyle data holds predictive power, its effectiveness in unsupervised settings may be limited without deeper feature representations or more refined techniques.

Although this study provides strong evidence supporting lifestyle-based DED prediction, several directions remain for future exploration. Improving the performance of unsupervised models. Incorporating limited clinical features in a hybrid framework may enhance model accuracy while maintaining a low-cost, non-invasive approach. Moreover, future versions of our system could integrate real-time user feedback to refine recommendations, adapt to changing habits, and support continuous monitoring. Expanding the model to include mobile health data or longitudinal lifestyle tracking would further enhance its practical value in everyday health management.

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Appendices

	Study Title	year	Objective	Data Type	Samples (N)	Algorithm Type	Performance score
1	Lifestyle and chronic kidney disease: A machine learning modeling study	2022	Predict risk of chronic kidney disease using only lifestyle and behavioral data	Lifestyle and Behavioral (self-reported)	470,778 participants	LightGBM (Gradient Boosting)	AUC: 0.710 (0.703–0.718), C-statistic: 0.706 (0.704–0.708)
2	An ensemble Machine Learning approach for predicting Type-II diabetes mellitus based on lifestyle indicators	2022	predict Type-II Diabetes Mellitus early using lifestyle indicators	Lifestyle Data	1939 patients	Bagged Decision Tree Random Forest AdaBoost Stochastic Gradient Boosting Voting Classifiers (Logistic Regression, Decision Tree, Support Vector Machine)	Bagged Decision Tree Accuracy: 99.41% Precision: 99.13% Recall: 95.83% Specificity: 99.11% F1-score: 99.15% MCR: 0.86% ROC: 99.07%

Appendix A — Table 2.1: Studies Using Lifestyle Data for Chronic Diseases

	Study Title	year	Objective	Data Type	Samples (N)	Algorithm Type	Performance score
3	A machine learning approach to predicting dry eye-related signs, symptoms and diagnoses from meibography images	2024	To identify relationships between Meibomian gland morphology and dry eye symptoms	infrared meibography images	562 images from 363 subjects (170 contact lens wearers, 193 non-wearers).	Deep Learning Logistic Regression	Accuracy for MG Morphological Characteristics: 65% to 99% Accuracy for Visible MG and Clinical Parameters: 74% to 85%
4	Machine Learning Models for the Diagnosis of Dry Eyes Using Real-world Clinical Data	2024	develop ML models for predicting the severity and type of DED using demographic and clinical data.	Real-world clinical data	1,313 samples	(Support Vector Machines - SVM).	Dry Eye Severity Model (SM): AUC-ROC: 0.79 AUC-PR: 0.61 Dry Eye Type Model (TM): AUC-ROC: 0.91 AUC-PR: 0.83
5	Classifying Dry Eye Disease Patients from Healthy Controls Using Machine Learning and Metabolomics Data	2024	classify cataract patients with and without dry eye disease	metabolomics data	81 samples (54 with dry eye, 27 without)	XGBoost Random Forest Support Vector Machine Multi-Layer Perceptron Logistic Regression K-Nearest Neighbor	Logistic Regression: AUC: 83.78% Balanced Accuracy: 73.5% MCC :0.5147 F1-Score: 85.13% Specificity: 56.67%
6	Impact of Incomplete Blinking Analyzed Using a Deep Learning Model With the Keratograph 5M in Dry Eye Disease	2023	Analyze blink patterns and assess dry eye disease (DED)	Blink Videos and images	100 samples (50 with dry eye, 50 without)	Deep Learning	Sensitivity: Higher at 30 FPS Average IPH: Higher in DED (P < 0.001) IB Frequency: Correlates with DED symptoms (R ≥ 0.195) IB Proportion: Higher in DED (R ≥ 0.202)
7	Interferometer eye image classification for dry eye categorization using phylogenetic diversity indexes for texture analysis	2020	Classify tear film patterns using images to support the diagnosis of dry eye syndrome.	Interferometric images of the tear film	106 images	Support Vector Machine (SVM), Random Forest (RF), Random Tree (RT), Naive Bayes (NB), Multilayer Perceptron (MLP)	RF: Accuracy: 97.54% Standard Deviation: 0.51% ROC Curve Area: 0.99 Kappa index: 0.96 F-Measure: 0.97
8	Dry eye is matched by increased intrasubject variability in tear osmolarity as confirmed by machine learning approach	2019	aim to assess tear osmolarity variability in dry eye patients as a potential diagnostic marker.	Clinical Measurements (Tear Osmolarity)	40 samples (20 with dry eye, 20 without)	Logistic Regression	Accuracy = 85%

Appendix B — Table 2.2: Studies Using Clinical Data for DED

	Study Title	year	Objective	Data Type	Samples (N)	Algorithm Type	Performance score
9	Artificial Intelligence Models Utilize Lifestyle Factors to Predict Dry Eye-Related Outcomes	2024	Predict dry eye symptoms and clinical signs using lifestyle and clinical data via ML	Clinical Lifestyle Questionnaire	363 subjects 726 records	Logistic Regression	Accuracy for Symptoms: 60.7% - 86.5% Accuracy for Diagnoses: 73.7% - 80.1% Accuracy for Clinical Signs: 66.9% - 98.7%
10	Explanatory Model of Dry Eye Disease Using Health and Nutrition Examinations: Machine Learning and Network-Based Factor Analysis from a National Survey	2020	aims to provide an integrative and personalized model for understanding Dry Eye Disease (DED) by analyzing multiple associated factors	Lifestyle and Health Examination Data	4,391 samples	Decision Trees Lasso Logistic Regression	AUROC = 0:70, 95% CI = 0:61 - 0:78, specificity = 68%, sensitivity= 66%

Appendix C — Table 2.3: Studies Combining Lifestyle and Clinical Data